INCH-POUND

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19 February 2026
SUPERSEDING
MIL-DTL-46594C
11 September 2011
MIL-DTL-46550D
w/AMENDMENT 1
12 September 2012

DETAIL SPECIFICATION

CHARGE, DEMOLITION: SHAPED, 40 LB, M3A1
PARTS, LOADING, ASSEMBLING AND PACKING

This specification is approved for use by all Departments and Agencies of the Department of War.

1. SCOPE

1.1 Scope. This specification covers the parts, loading, assembling and packing of the Charge, Demolition: Shaped, 40 LB, M3A1.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all requirements documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to: U.S. Army DEVCOM-AC, ATTN: FCDD-ACE-QSA , Picatinny Arsenal, New Jersey 07806-5000 or emailed to usarmy.pica.devcom-ac.list.ardec-stdzn-branch@army.mil. Since contact information can change, you may want to verify the currency of this information using ASSIST Online database at https://assist.dla.mil/.

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DEPARTMENT OF WAR STANDARDS

MIL-STD-1168	-	Ammunition Lot Numbering and Ammunition Data Card
MIL-STD-1916	-	DOD Preferred Method for Acceptance of Product

DEPARTMENT OF WAR SPECIFICATIONS

MIL-DTL-12560	-	Armor Plate, Steel, Wrought, Homogeneous (for use in combat-vehicles and for ammunition testing)

(Copies of these documents are available online at https://quicksearch.dla.mil/).

2.2.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation (see 6.2).

U.S. ARMY ARMAMENT CENTER DRAWINGS

7549165	-	Box, Packing, Ammunition, for Charge, Demolition: Shaped, 40 LB, M3A1
7549166	-	Support, Packing, Ammunition, for Charge, Shaped, 40 LB, M3A1
8858382	-	Charge, Demo: Shaped, 40 LB, M3A1, Metal Parts Assembly
8858386	-	Cone
8858390	-	Charge, Demolition: Shaped, 40 LB, M3A1
8858393	-	Booster
8858394	-	Charge, Demolition: Shaped, 40 LB, M3A1, Closing Cap Assembly
8858396	-	Cap, Closing
8858397	-	Detonator Well Assembly
8858398	-	Adapter, Well, Detonator
8858399	-	Well, Detonator
8858401	-	Stand-off Frame Assembly
13021902	-	Adapter, priming M1A5

(Copies of these drawings are available through email at usarmy.pica.devcom-ac.list.drawing-request-help-desk@army.mil.)

US ARMY DEFENSE AMMUNITION CENTER (DAC) DRAWINGS

19-48-4116/120D	-	Unitization Procedures for Boxed Ammunition and Components on 4-Way Entry Pallet

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(Copies of these documents may be requested online at https://www.dau.edu/cop/ammo/Pages/Default.aspx .)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM E1742/E1742M - Standard Practice for Radiographic Examination

(Copies of this document are available from https://www.astm.org/)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Conformance inspection. A sample shall be subjected to conformance inspection in accordance with 4.3.

3.3 Material, parts and assemblies. Materials, parts and assemblies shall comply with all requirements specified in drawings cited in section 2.2.2.

3.3.1 Specific gravity of booster. The specific gravity of the booster shall be not less than 1.50.

3.3.2 Average weight of explosive fill. The average weight of the explosive fill shall be not less than 29.1 lbs. (Minimum 29 lbs of Comp B plus 48 grams of booster material).

3.3.3 Radiographic examination. Cavitation shall be permitted only at the top side of the body where the explosive mixture is poured. Cavitation or voids in the loading assembly shall be not greater than ¼ inch depth and 1 inch width in any direction.

3.4 Leakproofness of closing cap assembly. The closing cap assembly shall not exhibit leakage when pressurized 2 to 3 psi internally with air for not less than 10 seconds.

3.5 Leakproofness of metal parts assembly. The metal parts assembly shall not exhibit leakage when pressurized 2 to 3 psi internally with air for not less than 10 seconds.

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3.6 Fit of closing cap assembly and metal parts assembly. The closing cap assembly and the metal parts assembly shall require a force fit to be inserted into the body assembly which does not leave the assembly loose or damaged.

3.7 Welds of stand-off frame assembly. The weld of each leg of the stand-off frame assembly shall withstand a free fall of not less than four (4) feet onto a steel or concrete surface, with the leg striking the surface at approximately thirty (30) degrees from a vertical position, without evidence of weld failure, either cracking, separating, or pulling apart.

3.8 Functioning test. The loaded M3A1 shall detonate high order with penetration of a target of Rolled Homogeneous Armor (RHA) in accordance with MIL-DTL-12560, Class 1 or 2, when initiated by a M6 or M7 blasting cap or equivalent military initiation device held in place with the priming adapter. The penetration hole in the plates shall exhibit no evidence of double- or multi-jets and have a minimum penetration of 4 inches. The entrance and exit of the penetration hole diameters of each plate will be two readings at 90 degrees apart for each diameter of the first 4 inches. Any penetration depth and diameter measurements beyond the minimum 4 inch plate thickness requirement will be for performance data collection purposes only.

3.9 Workmanship. All metal parts, assemblies and loading and packing materials shall be clean and free of explosive on exterior, chips, cracks, crazes, dirt, grease, rust, corrosion products, dents, burrs, sharp edges, surface defects, weld splatter and other defects and foreign matter. The cleaning method used shall not be injurious to any part or assembly nor shall the cleaning agent contaminate them. Exterior surface coatings shall be continuous except for a few light scratches not exposing the base material. All required markings shall be neat and sharply defined. Required packing shall be dry. Slight discoloration of the plunger and mold surfaces will be allowed, provide they do not indicate flaws or imperfections which would adversely affect the function of the case.

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4. VERIFICATION

TABLE I. Requirements/Verification cross reference matrix.

Method of Verification		Classes of Verification
1- Analysis 2- Examination 3- Test 4- Demonstration		A - First article B - Conformance

Section 3& 5 Requirements	Description	Verification Method				Verification Class		Section 4
		1	2	3	4	A	B	
3.1	First article	X	X	X		X		4.2
3.2	Conformance inspection			X			X	4.3
3.3	Material	X	X	X		X	X	4.3.3.1-4.3.3.15
3.3.1	Specific gravity of booster	X				X	X	4.3.4.1
3.3.2	Average weight of explosive fill		X			X	X	4.3.4.2
3.3.3	Radiographic examination		X			X	X	4.3.4.3
3.4	Leakproofness of cap assembly		X	X		X	X	4.3.4.5
3.5	Leakproofness of metal parts assembly		X	X		X	X	4.3.4.5
3.6	Fit of closing cap assembly and metal parts assembly		X			X	X	4.3.4.6
3.7	Welds of stand-off		X	X		X	X	4.3.4.4
3.8	Functioning test			X		X	X	4.3.4.7
3.9	Workmanship		X			X	X	4.3
5.1	Packaging and marking		X			X	X	4.3.3.14 4.3.3.15
5.2	Unitization		X				X	4.3.3.16

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4.1 Classification of Verification. The verification requirements specified herein are classified as follows:

a. First Article Inspection (see 4.2)

b. Conformance Inspection (see 4.3)

4.2 First article inspection. A sample shall be subjected to first article inspection in accordance with Table II. First article samples shall be manufactured using the same vendors, methods, materials, equipment, processes, procedures and facilities as will be used during regular production.

4.2.1 First article rejection. The first article sample shall be rejected if any of the inspections/tests fail to meet specification or drawing requirements. The Government reserves the right to terminate inspection or test upon failure of any sample to comply with any requirement.

4.2.2 Inspections to be performed. The first article samples shall be subjected to all of the examinations and tests specified in Table II.

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TABLE II. First article inspection.

Examination or test	Drawing(s)	Sample Size	Requirement paragraph	Inspection method
Well, detonator	8858399	25	3.3	4.3.3.1
Adapter, well, detonator	8858398	25	3.3	4.3.3.2
Detonator well assembly	8858397	25	3.3	4.3.3.3
Cap, closing	8858396	4.3.3.4	3.3	4.3.3.4
Closing, cap assembly Examination of defects Leakproofness of closing cap assembly	8858394	 25 3/ 100%	 3.3 3.4	 4.3.3.5 4.3.4.5
Cone	8858386	4.3.3.6	3.3	4.3.3.6
Body assembly, prior to painting and attaching carrying straps Examination of defects Leakproofness of closing metal parts assembly Fit of closing cap assembly and metal parts assembly	8858382	 25 3/ 100% 25	 3.3 3.5 3.6	 4.3.3.7 4.3.4.5 4.3.4.6
Body assembly	8858382	25	3.3	4.3.3.8
Welds of stand-off frame assembly Examination of defects Weld stand-off	8858401	 4.3.3.9 25	 3.3 3.7	 4.3.3.9 4.3.4.4
Booster Examination of defects Specific gravity	8858393	 25 10	 3.3 3.3.1	 4.3.3.10 4.3.4.1
Assembly prior to loading	8858390	25	3.3	4.3.3.11
Assembly prior to closing	8858390	25	3.3	4.3.3.12
Loading and marking assembly Examination of defects Radiographic examination Average weight Function test	8858390	 25 2/ 10 25 2/ 10 1/	 3.3 3.3.3 3.3.2 3.8	 4.3.3.13 4.3.4.3 4.3.4.2 4.3.4.7
Box , packing, prior to closing	7549165 or 7549166	4.3.3.14 2/	5.1	4.3..3.15
Box, packing, sealed	7549165	4.3.3.15 2/	5.1	4.3.3.15
Unitization	19-48-4116 19-48-4116/120H	3 pallot loads	5.2	4.3.3.16

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TABLE II. First article inspection (continued).

NOTE:
1/ Radiographic examination shall be conducted on the First Article samples that are to be subjected to the Functioning Test, prior to the execution of the Functioning Test. Any test unit which fails to detonate high order shall be classified as defective.
2/ Certificate of analysis shall be provided for all materials.
3/ 100% inspection of all drawing characteristics shall be performed.

4.3 Conformance inspection.

4.3.1 Inspection lot formation. Inspection lots shall conform to the requirements of MIL-STD-1916. Inspection lots shall contain materials from not more than one lot interfix number from one manufacturer (when applicable). When not applicable, lot formation shall be homogeneous. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings and revisions, and the same specifications and revisions. In addition, each lot shall contain materials from not more than one lot interfix number from one manufacturer.

4.3.2 Ammunition lot numbering.. Visually verify that a unique ammunition lot number has been assigned to each lot as described in MIL-STD-1168.

4.3.3 Inspection requirements by classification of characteristics.

a. For the conformance inspection paragraphs 4.3.3.1 through 4.3.3.16, the definitions of critical, major, and minor defects are provided in paragraph “Definitions” of MIL-STD-1916.
b. The acceptance criteria for sampling inspection shall be in accordance with the levels provided in conformance examination/test paragraph and MIL-STD-1916.
c. Alternative conformance acceptance. Alternate conformance procedures may be proposed (See paragraph “Acceptance by contractor-proposed provisions” of MIL-STD-1916).

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4.3.3.1 Well, detonator.				Drawing number 8858399 Next higher assembly 8858397
	Examination or test	Conformance criteria	Requirement paragraph	Inspection Method
Critical	None defined			
Major				
101	Diameter and length of cavity	VL IV	3.3	Gage
102	Outside diameter of body, maximum (max.)	VL IV	3.3	Gage
103	Wall thickness, minimum (min.)	VL IV	3.3	Gage
Minor				
201	Diameter of flange, max.	VL II	3.3	Gage
202	Thickness of flange, max.	VL II	3.3	Gage
203	Length of closed end, max.	VL II	3.3	Gage
204	Evidence of poor workmanship	VL II	3.9	Visual

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4.3.3.2 Adaptor, well, detonator.				Drawing number 8858398 Next higher assembly 8858397
	Examination or test	Conformance criteria	Requirement paragraph	Inspection Method
Critical	None defined			
Major				
101	Pitch diameter of thread, min.	VL IV	3.3	Gage
102	Minor diameter of thread, max.	VL IV	3.3	Gage
Minor				
201	Overall length, min.	VL II	3.3	Gage
202	Evidence of poor workmanship	VL II	3.9	Visual

4.3.3.3 Detonator well assembly.				Drawing number 8858397 Next higher assembly 8858394
	Examination or test	Conformance criteria	Requirement paragraph	Inspection Method
Critical	None defined			
Major				
101	Concentricity of inside diameter with thread	VL IV	3.3	Gage
Minor				
201	Assembled length of adaptor, min.	VL II	3.3	Gage
202	Evidence of poor workmanship	VL II	3.9	Visual

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4.3.3.4 Cap, closing.				Drawing number 8858396 Next higher assembly 8858394
	Examination or test	Conformance criteria	Requirement paragraph	Inspection Method
Critical	None defined			
Major	None defined			
Minor				
201	Inside depth, min.	VL II	3.3	Gage
202	Flange improperly formed	VL II	3.3	Visual
203	Evidence of poor workmanship	VL II	3.9	Visual

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4.3.3.5 Closing cap assembly.				Drawing number 8858394 Next higher assembly 8858382
	Examination or test	Conformance criteria	Requirement paragraph	Inspection Method
Critical	None defined			
Major				
101	Concentricity of inside diameter of closing cap to thread	VL IV	3.3	Gage
102	Dent in thread or detonator cavity	VL IV	3.3	Visual
103	Tape missing, loose, or mislocated	VL IV	3.3	Visual
104	Enamel paint or color improper	VL IV	3.3	Visual
105	Leakproofness of closing cap assembly	100%	3.4	4.3.4.5
Minor				
201	Length and diameter of cavity inside	VL II	3.3	Gage
202	Paint in detonator well	VL II	3.3	Visual
203	Protective coating missing, inadequate, or improper	VL II	3.3	Visual
204	Evidence of poor workmanship	VL II	3.9	Visual

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4.3.3.6 Cone.				Drawing number 8858386 Next higher assembly 8858382
	Examination or test	Conformance criteria	Requirement paragraph	Inspection Method
Critical	None defined			
Major				
101	Concentricity of lip with either datum diameter	VL IV	3.3	Gage
102	Variation in wall thickness	VL IV	3.3	Gage
Minor				
201	Diameter at lip	VL II	3.3	Gage
202	Wall thickness	VL II	3.3	Gage
203	Total height, max.	VL II	3.3	Gage
204	Evidence of poor workmanship	VL II	3.9	Visual

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4.3.3.7 Metal parts assembly, prior to painting.				Drawing number 8858382 Next higher assembly 8858390
	Examination or test	Conformance criteria	Requirement paragraph	Inspection Method
Critical	None defined			
Major				
101	Concentricity of container neck with either datum diameter on cone	VL IV	3.3	Gage
102	Total length	VL IV	3.3	Gage
103	Sealer missing or improper at junction of cone and body	VL IV	3.3	Visual
104	Cone loose or crimp inadequate	VL IV	3.3	Visual/Manual
105	Bracket weld missing or inadequate	VL IV	3.3	Visual/Manual
106	Leakproofness of body assembly	100%	3.5	4.3.4.5
107	Fit of closing cap assembly	25	3.6	4.3.4.6
Minor				
201	Outside diameter, max.	VL II	3.3	Gage
202	Evidence of poor workmanship	VL II	3.9	Visual

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4.3.3.8 Metal parts assembly.				Drawing number 8858382 Next higher assembly 8858390
	Examination or test	Conformance criteria	Requirement paragraph	Inspection Method
Critical	None defined			
Major				
101	Bare spot on interior surface	VL IV	3.3	Visual
102	Paint improper or inadequate	VL IV	3.3	Visual
103	Foreign matter in interior	VL IV	3.3	Visual
Minor				
201	Carrying strap missing, broken, improperly assembled, or insecure	VL II	3.3	Visual/Manual
202	Evidence of poor workmanship	VL II	3.9	Visual

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4.3.3.9 Stand-off frame assembly.				Drawing number 8858401 Next higher assembly 8858382
	Examination or test	Conformance criteria	Requirement paragraph	Inspection Method
Critical	None defined			
Major				
101	Total length, max.	VL IV	3.3	Gage
102	Thumb screw missing	VL IV	3.3	Visual
103	Welds of stand-off 1/	25	3.7	4.3.4.4
Minor				
201	Protective coating missing or inadequate	VL II	3.3	Visual
202	Evidence of poor workmanship	VL II	3.9	Visual
NOTE: 1/ If any sample exhibits evidence of bending or deformation in the stand-off legs, the sample shall be rejected.				

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4.3.3.10 Booster.				Drawing number 8858393 Next higher assembly 8858390
	Examination or test	Conformance criteria	Requirement paragraph	Inspection Method
Critical	None defined			
Major				
101	Concentricity of small cavity to outside diameter	VL IV	3.3	Gage
102	Diameter of small cavity, minimum (min.)	VL IV	3.3	Gage
103	Depth of large cavity, min.	VL IV	3.3	Gage
104	Specific gravity	4.3.4.1	3.3	4.3.4.1
Minor				
201	Depth of small cavity, min.	VL II	3.3	Gage
202	Diameter of large cavity, min.	VL II	3.3	Gage
203	Total length, maximum (max.)	VL II	3.3	Gage
204	Radius missing	VL II	3.3	Visual
205	Evidence of poor workmanship	VL II	3.13	Visual

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4.3.3.11 Assembly, Prior to Loading.				Drawing number 8858390 Next higher assembly 7549165 or 7549166
	Examination or test	Conformance criteria	Requirement paragraph	Inspection Method
Critical	None Defined			
Major				
101	Interior coating with bare spot	VL IV	3.3	Visual
102	Foreign matter in interior	VL IV	3.3	Visual
Minor				
201	Evidence of poor workmanship	VL II	3.13	Visual

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4.3.3.12 Assembly, prior to closing.				Drawing number 8858390 Next higher assembly 7549165 or 7549166
	Examination or test	Conformance criteria	Requirement paragraph	Inspection Method
Critical	None Defined			
Major				
101	Charge improper 1/	VL IV	3.3	Visual
102	Booster missing	VL IV	3.3	Visual
103	Pad or spacer missing (M3A1 only)	VL IV	3.3	Visual
104	Container damaged	VL IV	3.3	Visual
Minor				
201	Depth to charge	VL II	3.3	Gage
202	Evidence of poor workmanship	VL II	3.13	Visual
NOTE: 1/ Charge misaligned with detonator well or not the correct weight/density				

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4.3.3.13 Charge, Demolition: Shaped, 40 LB, M3A1.				Drawing number 8858390 Next higher assembly 7549165 or 7549166
	Examination or test	Conformance criteria	Requirement paragraph	Inspection Method
Critical	None defined			
Major				
101	Gage does not fully enter detonator well	VL IV	3.3	Gage
102	Closing cap assembly not tight or adhesive missing	VL IV	3.3	Visual/Manual
103	Protective tape missing	VL IV	3.3	Visual
104	Component missing	VL IV	3.3	Visual
105	Strap slide improperly assembled	VL IV	3.3	Visual
106	Weight of charge	4.3.4.2	3.3.2	4.3.4.2
107	Radiographic examination	10	3.3.3	4.3.4.3
108	Functioning Test	10 1/	3.8	4.3.4.7
Minor				
201	Total length, max.	VL II	3.3	Gage
202	Width at stand-off legs, max	VL II	3.3	Gage
203	Marking missing, incorrect, illegible, or not in proper location	VL II	3.9	Visual
204	Evidence of poor workmanship	VL II	3.9	Visual

NOTE:
1/ After three (3) consecutive lots have met the criteria, a sample of five (5) test units shall be selected and function tested out of each lot produced. If one or more loaded units fail to function (or a new First Article is required), the sampling shall return to the original test plan and continue until three (3) consecutive lots have complied with the acceptance criteria specified.

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4.3.3.14 Box, packing, prior to closing.				Drawing number 7549165 or 7549166 Next higher assembly 19-48-4116 19-48-4116/120D
	Examination or test	Conformance criteria	Requirement paragraph	Inspection Method
Critical	None defined			
Major				
101	Support missing	VL II	5.1	Visual
102	Packing component missing	VL II	5.1	Visual/Manual
Minor	None defined			
201	Presence of M1A5 adapters (Dwg #13021902)	VL I	5.1	Visual
202	Evidence of poor workmanship	VL I	5.1	Visual

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4.3.3.15 Box, packing, sealed.				Drawing number 7549165 or 7549166 Next higher assembly 19-48-4116 19-48-4116/120D
	Examination or test	Conformance criteria	Requirement paragraph	Inspection Method
Critical	None defined			
Major				
101	Contents of box exposed	VL IV	5.1	Visual
102	Board broken	VL IV	5.1	Visual
103	NSN number missing or misleading	VL IV	5.1	Visual
104	Hardware missing, broken or loose	VL IV	5.1	Visual
105	Strapping missing, broken or loose	VL IV	5.1	Visual
106	Contents loose when box is inverted	VL IV	5.1	Visual/Manual
Minor				
201	Marking missing, incorrect, illegible, or not in proper location	VL II	5.1	Visual
202	Hardware improperly engaged	VL II	5.1	Visual/Manual
203	Strapping improperly engaged	VL II	5.1	Visual/Manual
204	Metallic seal missing, missing manufacturer's identification, incorrect, not secured, or not applied correctly	VL II	5.1	Visual/Manual
205	Car seal missing, unsealed or improperly positioned	VL II	5.1	Visual/Manual
206	Handle missing or insecure	VL II	5.1	Visual/Manual

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4.3.3.16 Unitization.			Drawing Number 19-48-4116 19-48-4116/120D Next Higher Assembly 13025743	
	Examination or test	Conformance Criteria	Requirement Paragraph	Inspection Method
Critical	None defined			
Major				
101	Wood preservative and heat treatment markings for the pallet missing, incorrect, illegible, or not in proper location	VL II	5.2	Visual
102	Pallet load markings -including 2D bar codes - missing, incorrect, illegible, or not in proper location	VL II	5.2	Visual
103	Filler assembly not built, treated, or marked correctly	VL II	5.2	Visual
104	Boxes incorrectly oriented or positioned on pallet	VL II	5.2	Visual
105	Strapping not applied correctly or incorrect type	VL II	5.2	Visual
106	Unit load not tightly strapped or any strap loose	VL II	5.2	Visual / Manual
Minor				
201	Workmanship	VL I	5.1	Visual

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4.3.4 Test methods and procedures.

4.3.4.1 Specific gravity of booster. A sample of ten (10) boosters shall be selected from each lot for this test. If any booster fails to meet the requirement specified, the lot shall be rejected. The booster shall be weighed on a balance accurate to 0.01 grams and the weight recorded as dry weight. The booster shall then be weighed while completely submerged in water, taking care to remove all air bubbles inside the detonator well and remaining on the surfaces of the booster. The temperature of the distilled water shall be 68 degrees ºF +/- 2 ºF. This weight shall be recorded as weight under water. (When submerged, the position of the booster shall be such that no air is entrapped.) Specific gravity shall be calculated as follows:

$$\textit{Specific gravity} = \frac{\textit{Weight of booster dry}}{\textit{(Weight of booster dry) minus (Weight of booster immersed in water)}} \times \textit{Specific gravity of } H_2O \textit{ at noted temperature (see Table III).}$$

TABLE III. Specific gravity of water within temperature range.

Temp °F	Temp °C	Specific Gravity of H_2O
66	18.89	0.9984
67	19.44	0.9983
68	20.00	0.9982
69	20.56	0.9981
70	21.11	0.9980

4.3.4.2 Average weight of explosive fill. A sample of 25 shaped charges shall be selected from each lot and weighed on a scale capable of determining weights within an accuracy of two (2) ounces. The average weight of 30 metal parts assemblies shall be subtracted from each individual weight to determine compliance with the requirement. Any unit failing to meet the weight requirements shall be classed defective and the lot shall be rejected.

4.3.4.3 Radiographic examination. Radiographic equipment, operations and procedures shall be in accordance with ASTM E1742/E1742M (see 6.5). The radiographic examination shall detect voids, cracks, cavitation, separation, and cracking of the explosive and presence of any foreign material. The assembly shall be radiographically inspected two (2) times. The assembly shall be inspected along its length, rotated approximately ninety degrees (90º) on its cylindrical axis, and inspected a second time. Any assembly which does not meet the criteria shall reject the sample(s).

4.3.4.4 Welds of stand-off frame assembly. Unless otherwise noted, a sample of twenty-five (25) stand-off frame assemblies shall be selected from each lot for this test. Verify by test that there are no weld failures. Presence of a weld failure shall reject the lot.

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4.3.4.5 Leakproofness of closing cap and metal parts assembly. The required air pressure shall be applied to both the closing cap and metal parts assemblies, and both assemblies shall then be completely submerged in water individually. Observations shall be made for a stream or streams of bubbles, of any frequency, indicating leakage in the assembly. Any unit exhibiting leakage shall be classed defective and removed from the lot. Units that pass the leakproofness test shall be thoroughly dried and returned to the lot.

4.3.4.6 Fit of closing cap assembly and metal parts assembly. Unless otherwise noted, a sample of twenty-five (25) closing cap assemblies and twenty-five (25) metal parts assemblies shall be selected from each lot for this test. The samples of closing cap assemblies and metal parts assemblies shall be segregated into combinations consisting of one unit of each. The closing cap assembly shall be assembled to its metal parts assembly using applicable equipment. Failure to require a force fit shall class a combination defective. Failure of one or more combinations in meeting the requirement shall reject the lot. Any unit failing to meet the requirement shall be classed defective. Units that pass the leakproofness test shall be thoroughly dried and returned to the lot.

4.3.4.7 Functioning test. Ten (10) test units shall be selected from each lot. The loaded assembly shall be assembled for firing and positioned on the steel plate. The test shall be detonated using an M6 or M7 blasting cap (see 6.2i) and shall be secured in the detonator well with the priming adapter and the unit initiated. The test shall also be conducted at local temperatures and temperatures shall be recorded for data collection purposes only. Any test unit which fails to detonate high order shall be classified as defective and the lot is to be rejected. Observation shall be made for the presence of undetonated charge and for evidence of double- or multi-jets. Presence of either shall classify a unit defective. The RHA target shall be of a size not less than twenty (20) inches in total thickness with a maximum of ten (10) plates making up that thickness. The first four (4) inches shall be made up of a maximum of two (2) plates at two (2) inches minimum thickness each. The surface area shall not be less than 144 inches made up with one (1) plate. Any penetration depth not meeting the requirements shall be classified as a unit failure and the lot is to be rejected. After three (3) consecutive lots have met the criteria, a sample of five (5) test units shall be selected and function tested out of each lot produced. If one or more loaded units fail to function, the lot shall be rejected and the sampling shall return to the original test plan.

5. PACKAGING

5.1 Packaging and marking. The materials, components, and assemblies shall comply with Drawing 7549165 for the M3A1 and all their subsidiary drawings.

5.2 Unitization. Palletize M3A1 in accordance with 19-48-4116/120D and their subsidiary drawings.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory).

6.1 Intended use. The 40 lb. Shaped Demolition Charge is primarily employed for burring holes in earth, metal, concrete, and paved or unpaved roads

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Provisions for submission of first article samples.

c. Requirements for Ammunition Lot Numbering, if applicable.

d. Requirement for propellant and energetic materials IAW MIL-STD-1171

e. Requirement for acceptance inspection equipment (AIE) designs.

f. Certificate of Conformance for each lot or shipment of product.

g. Preservation, packaging, and marking.

h. Radiographic personnel should be qualified in accordance with NAS 410.

i. The 40 lb. Shaped Demolition Charge is functioned with either an M6 Blasting Cap in accordance with MIL-PRF-32293A or an M7 Blasting Cap in accordance with MIL-DTL-45469F and Drawing 8830948.

6.3 Submission of inspection equipment designs for approval. Equipment designs should be submitted per the Contract Data Requirements Lists (DD Form 1423).

6.4 Acceptance inspection equipment. Complete results of the acceptance inspection equipment should be available to the Government through the Contracting Officer. Distribution should include: DEVCOM-AC, ATTN: FCDD-ACE-QWE , Picatinny, NJ 07806-5000.

6.5 Radiographic examination. Complete results of the radiographic examination should be available to the Government through the Contracting Officer. Distribution should include: U.S. Army DEVCOM-AC, FCDD-ACE-QWE Picatinny, NJ 07806-5000.

a. National Aerospace Standard

NAS 410 - NAS Certification and Qualification of Nondestructive Test Personnel

(Copies of this document are available from https://www.aia-aerospace.org/ or Aerospace

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Industries Association, 1000 Wilson Boulevard, Suite 1700, Arlington, VA 22209-3928.)

6.6 Certification. Unless otherwise specified by the Contracting Officer, certification should be submitted by the contractor in accordance with the contract. Distribution should include: U.S. Army DEVCOM-AC, FCDD-ACE-QWE Picatinny, NJ 07806-5000.

6.6 Subject term (key word) listing.

Explosive
Composition B
Detonation
Kit
Munitions

6.7 Superseding explanatory note. This revision combines MIL-DTL-46594C with MIL-DTL-46550D w/Amendment 1 into a single specification. MIL-DTL-46550D covered the metal parts assembly for manufacturing the Charge, Demolition, Shaped, 40 Lb. M3A1. As a result of the combination of the two specifications, MIL-DTL-46550D has been cancelled.

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIALS

Custodian:
Army-AR
Navy -OS
Air Force – 70

Preparing Activity:
Army-AR
(Project 1375-2026-002)

Review Activity
Navy – AS
Air Force – 11

Civil agency:
GSA-FAS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

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